Group I, Row III ($\Lambda \to 0$, $\vee \to \infty$): simplification of $N_2 + N_{3a}$

Step 0: notation and common factor

Every term has the same prefactor, phase and colour structure:

$$\frac{1}{(2\pi)^3}\,\frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{w,w'} e^{-ik\cdot(w'-w)}\int_{x,y,z,x'}\frac{1}{(x'-w')^2(y-z)^2}\,(\ldots)\,\mathcal{C}, \tag{1}$$

$$\mathcal{C} = \Big[U^{ab'}(x')\rho^{b'}(x') - U^{ab'}(w')\rho^{b'}(x')\Big]\left[U^{de}(x)U^{bc}(z)\rho^e(x)\rho^c(y) - U^{de}(x)U^{bc}(y)\rho^e(x)\rho^c(y)\right]. \tag{2}$$

Below we write only $(\ldots)$. We abbreviate

$$X = x - w, \quad Z = x - z, \quad W = z - w = X - Z, \quad X' = x' - w', \quad Y = y - z, \quad D = X^2 - Z^2, \quad u = \frac{Z^2}{X^2}, \tag{3}$$

$$L_2 = \log\frac{-(x-z)^2}{(x-w)^2-(x-z)^2} = \log\frac{-Z^2}{D}, \qquad L_3 = \log\frac{(x-w)^2-(x-z)^2}{(x-w)^2} = \log\frac{D}{X^2}. \tag{4}$$

Step 1: the terms

N_2 has an overall minus sign in front of its braces. Taking it inside, $N_2 = L_2\,(a_1 + \cdots + a_6)$ with

$$a_1 = -\frac{(X'\cdot W)(Y\cdot X)}{W^2 D}, \qquad a_2 = +\frac{(X'\cdot X)(Y\cdot W)\,Z^2}{W^2 D\,X^2}, \qquad a_3 = -\frac{(X'\cdot Y)(W\cdot X)\,Z^2}{W^2 D^2},$$
$$a_4 = +\frac{(X'\cdot W)(Y\cdot Z)\,X^2}{W^2 D\,Z^2}, \qquad a_5 = -\frac{(X'\cdot Z)(Y\cdot W)}{W^2 D}, \qquad a_6 = +\frac{(X'\cdot Y)(W\cdot Z)\,X^2}{W^2 D^2}. \tag{5}$$

$N_{3a} = L_3\,(b_1 + \cdots + b_6) + \mathcal{A}_0\,(c_1 + c_2 + c_3)$ with

$$b_1 = -\frac{(X'\cdot Y)(W\cdot X)\,Z^2}{W^2 D^2}, \qquad b_2 = +\frac{(X'\cdot Y)(W\cdot Z)\,X^2}{W^2 D^2}, \qquad b_3 = -\frac{(X'\cdot W)(Y\cdot X)}{W^2 D},$$
$$b_4 = +\frac{(X'\cdot W)(Y\cdot Z)\,X^2}{W^2 D\,Z^2}, \qquad b_5 = +\frac{(Y\cdot W)(X'\cdot X)\,Z^2}{W^2 D\,X^2}, \qquad b_6 = -\frac{(Y\cdot W)(X'\cdot Z)}{W^2 D}, \tag{6}$$

$$c_1 = +\frac{(X'\cdot W)(Y\cdot X)}{W^2}, \qquad c_2 = +\frac{(Y\cdot W)(X'\cdot Z)}{W^2}, \qquad c_3 = -\frac{(X\cdot Z)(X'\cdot W)(Y\cdot Z)}{W^2 Z^2}. \tag{7}$$

Step 2: the N_2 and N_{3a} tensors are the same

Comparing term by term:

$$a_1 = b_3, \quad a_2 = b_5, \quad a_3 = b_1, \quad a_4 = b_4, \quad a_5 = b_6, \quad a_6 = b_2. \tag{8}$$

Hence, with $\mathcal{T} \equiv b_1 + \cdots + b_6$,

$$N_2 + N_{3a}\big|_{\log} = (L_2 + L_3)\,\mathcal{T}. \tag{9}$$

Step 3: adding the two logarithms

Both logarithms come from the same energy denominator and are on the principal branch, $\log(-A) = \log A + i\pi$ for $A > 0$.

- $u < 1$ ($D > 0$): $L_2 = \log\frac{u}{1-u} + i\pi$ and $L_3 = \log(1-u)$.
- $u > 1$ ($D < 0$): $L_2 = \log\frac{u}{u-1}$ and $L_3 = \log(u-1) + i\pi$.

In both cases

$$L_2 + L_3 = \log\frac{-Z^2}{X^2} = \log\frac{(x-z)^2}{(x-w)^2} + i\pi. \tag{10}$$

The logarithms $\log|1-u|$, which are singular at $|x-z| = |x-w|$, have cancelled.

Step 4: reducing the tensor $\mathcal{T}$

Write $\mathcal{T} = X'^i Y^j \mathcal{T}^{ij}$. From Step 1,

$$W^2 D\, \mathcal{T}^{ij} = \underbrace{-W^i X^j + u\, X^i W^j + \frac{1}{u}\, W^i Z^j - Z^i W^j}_{\text{non-}\delta\ (b_3,b_5,b_4,b_6)} + \underbrace{\delta^{ij}\, \frac{(W\cdot Z)X^2 - (W\cdot X)Z^2}{D}}_{\delta\ (b_1,b_2)}. \tag{11}$$

Non-δ part. Use $X = W + Z$ in the first two terms:

$$\begin{aligned} -W^i X^j + uX^i W^j &= -W^i W^j - W^i Z^j + u\, W^i W^j + u\, Z^i W^j, \\ \Rightarrow\ \text{non-}\delta &= (u-1)W^i W^j + (u-1)Z^i W^j + \Big(\frac{1}{u} - 1\Big) W^i Z^j = (u-1)\Big[X^i W^j - \frac{1}{u} W^i Z^j\Big]. \end{aligned} \tag{12}$$

Since $D = X^2(1-u)$, the factor $(u-1)$ cancels the pole:

$$\frac{\text{non-}\delta}{W^2 D} = -\frac{X^i W^j}{W^2 X^2} + \frac{W^i Z^j}{W^2 Z^2}. \tag{13}$$

δ part. Use $W = X - Z$:

$$(W\cdot Z)X^2 - (W\cdot X)Z^2 = (X\cdot Z)(X^2 + Z^2) - 2X^2 Z^2. \tag{14}$$

Then use $2\, X\cdot Z = X^2 + Z^2 - W^2$:

$$(X\cdot Z)(X^2+Z^2) - 2X^2Z^2 = \tfrac{1}{2}\big[(X^2+Z^2)^2 - 4X^2Z^2\big] - \tfrac{1}{2}W^2(X^2+Z^2) = \tfrac{1}{2}\big[D^2 - W^2(X^2+Z^2)\big]. \tag{15}$$

Dividing by $W^2 D^2$:

$$\frac{\delta\text{ part}}{W^2 D} = \delta^{ij}\Big[\frac{1}{2W^2} - \frac{X^2+Z^2}{2D^2}\Big]. \tag{16}$$

Result.

$$\mathcal{T}^{ij} = \frac{\delta^{ij}}{2(z-w)^2} - \frac{(x-w)^i (z-w)^j}{(z-w)^2 (x-w)^2} + \frac{(z-w)^i (x-z)^j}{(z-w)^2 (x-z)^2} - \frac{\delta^{ij}\left[(x-w)^2 + (x-z)^2\right]}{2\big[(x-w)^2 - (x-z)^2\big]^2}. \tag{17}$$

In terms of the Row IV kernel $\mathcal{K}^{ij}_{\rm IV} = \frac{\delta^{ij}}{2W^2} + \frac{W^i Z^j}{W^2 Z^2} - \frac{X^i W^j}{W^2 X^2} - \frac{X^i Z^j}{X^2 Z^2}$,

$$\mathcal{T}^{ij} = \mathcal{K}^{ij}_{\rm IV} + \frac{X^i Z^j}{X^2 Z^2} - \frac{\delta^{ij}(X^2+Z^2)}{2D^2}. \tag{18}$$

Step 5: the $\mathcal{A}_0$ terms

c_1 and c_3 share $(X'\cdot W)$:

$$c_1 + c_3 = \frac{(X'\cdot W)}{W^2}\Big[Y\cdot X - \frac{(X\cdot Z)(Y\cdot Z)}{Z^2}\Big] = \frac{(X'\cdot W)\,(Y\cdot X_\perp)}{W^2}, \qquad X_\perp \equiv X - \frac{X\cdot Z}{Z^2}\, Z. \tag{19}$$

$X_\perp$ is the component of $x-w$ perpendicular to $x-z$. In geometric form, $\mathcal{A}_0 = \frac{\theta_w}{|(x-w)\times(z-w)|}$, where $\theta_w = \arccos\frac{(x-w)\cdot(z-w)}{|x-w||z-w|}$ is the angle of the triangle (x,z,w) at w. So

$$\mathcal{A}_0(c_1+c_2+c_3) = \frac{\theta_w}{|(x-w)\times(z-w)|\,(z-w)^2}\Big[(x'-w')\cdot(z-w)\ (y-z)\cdot X_\perp + (x'-w')\cdot(x-z)\ (y-z)\cdot(z-w)\Big]. \tag{20}$$

Step 6: final result

$$\begin{aligned} \frac{d^3 N_{2\,\rm NLO}}{d^3k} + \frac{d^3 N_{3a\,\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{w,w'} e^{-ik\cdot(w'-w)} \int_{x,y,z,x'} \frac{1}{(x'-w')^2(y-z)^2}\Bigg\{ \\ \Big[\log\frac{(x-z)^2}{(x-w)^2} + i\pi\Big](x'-w')^i (y-z)^j \Bigg[\frac{\delta^{ij}}{2(z-w)^2} - \frac{(x-w)^i(z-w)^j}{(z-w)^2(x-w)^2} \\ + \frac{(z-w)^i(x-z)^j}{(z-w)^2(x-z)^2} - \frac{\delta^{ij}\left[(x-w)^2+(x-z)^2\right]}{2\big[(x-w)^2-(x-z)^2\big]^2}\Bigg] \\ + \frac{\theta_w}{|(x-w)\times(z-w)|\,(z-w)^2}\Big[(x'-w')\cdot(z-w)\ (y-z)\cdot X_\perp + (x'-w')\cdot(x-z)\ (y-z)\cdot(z-w)\Big]\Bigg\} \\ \times\Big[U^{ab'}(x')\rho^{b'}(x') - U^{ab'}(w')\rho^{b'}(x')\Big]\left[U^{de}(x)U^{bc}(z)\rho^e(x)\rho^c(y) - U^{de}(x)U^{bc}(y)\rho^e(x)\rho^c(y)\right]. \end{aligned} \tag{21}$$

This was checked numerically against the original $6+6+3$ terms at random points, for $u<1$ and $u>1$. The agreement is 10^{-14}.

Remarks

- **Reality.** $i\big[\log u + i\pi\big] = i\log u - \pi$. The $i\log u$ part and the $\mathcal{A}_0$ part give $\sin(k\cdot(w'-w))$ with the c.c. The $-\pi$ part gives a real term, which becomes $\cos(k\cdot(w'-w))$.

- **The remaining pole.** The last δ^{ij} term has a double pole at $(x-z)^2=(x-w)^2$. The $i\pi$ in front does not vanish there, so N_2+N_{3a} is *not* integrable by itself. The pole cancels against N_1. Its two log terms with kernel $\frac{(X^2+Z^2)\delta^{ij}}{(Z^2-X^2)^2}$ add up to $\log u + i\pi$, and its third term is $-2\frac{X^iZ^j}{X^2Z^2}\log\frac{-Z^2}{X^2}$. In the normalisation $\frac{ig^4f^{abd}}{8\pi^5}$ above, N_1 is

$$(\log u + i\pi)\Big[\frac{\delta^{ij}(X^2+Z^2)}{2D^2} - \frac{X^iZ^j}{X^2Z^2}\Big], \tag{22}$$

  which removes exactly the two extra pieces in $\mathcal{T}^{ij} = \mathcal{K}^{ij}_{\rm IV} + \frac{X^iZ^j}{X^2Z^2} - \frac{\delta^{ij}(X^2+Z^2)}{2D^2}$. Therefore

$$N_1+N_2+N_{3a} = \Big[\log\frac{(x-z)^2}{(x-w)^2} + i\pi\Big]\,(x'-w')^i(y-z)^j\,\mathcal{K}^{ij}_{\rm IV} \;+\; \mathcal{A}_0 \text{ terms}, \tag{23}$$

  which is free of all $\frac{1}{(x-w)^2-(x-z)^2}$ poles. This agrees with the Row III result in Limiting_Cases_updated.